

Domain Control and Active Directory Lab

Part 2 – Networking

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Technologies Used

- **Oracle VirtualBox**
- **Windows Server 2022**
- **Network Adapters Connections**
- **Server Manager**

OBJECTIVE:

The objective of this lab is to configure the Domain Controller's external and internal network adapters, establishing Internet connectivity and a private enterprise network that future Windows clients will use for DNS, DHCP, and Active Directory services.

Lab Setup :

This lab was built using Oracle VirtualBox to simulate a small enterprise Windows Server environment. The environment consists of a Windows Server 2022 virtual machine that will function as the Domain Controller and provide centralized network services for future Windows client systems. During this phase, the server was configured with two virtual network adapters to separate Internet connectivity from the internal enterprise network. The external adapter utilizes VirtualBox's built-in NAT network for Internet access, while the internal adapter was configured with a static IPv4 address (172.24.8.1/24) to establish a private network for future Active Directory, DNS and DHCP services. The Domain Controller was also configured to use the loopback address (127.0.0.1) as its preferred DNS server, allowing the server to reference its own DNS service once installed. This networking configuration provides a stable foundation for promoting the server to a Domain Controller and allows future Windows clients to reliably communicate with centralized network services.

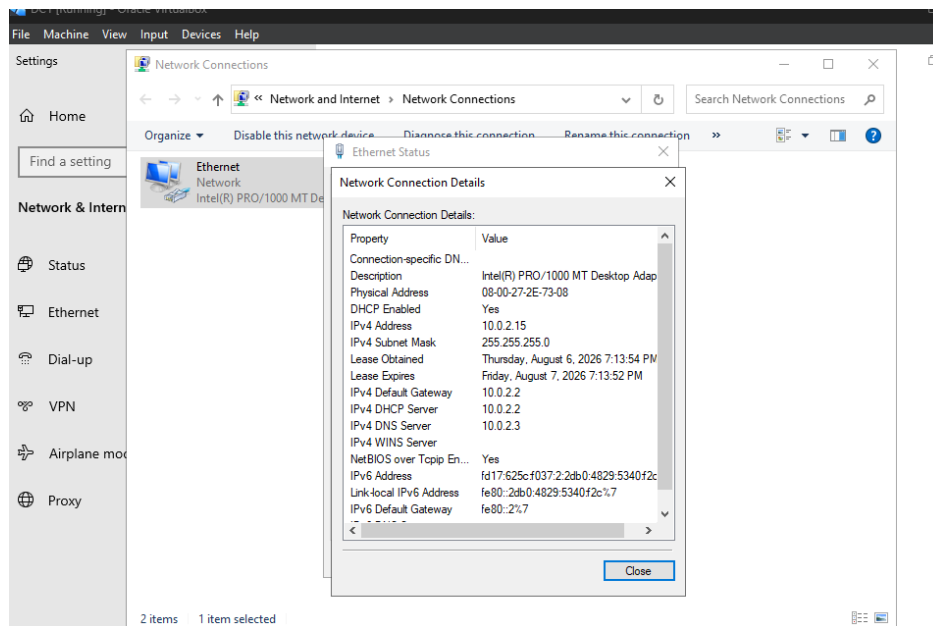


Figure 1. NAT connects to external host network, by default VirtualBox assigns 10.0.2.x IP address to “Ethernet” Connection. External adapters allow server to gain internet access for updates, software updates, etc.

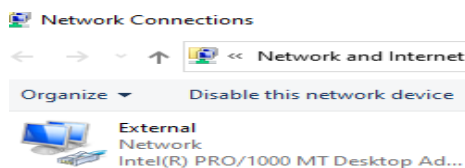


Figure 2. Ethernet connection renamed to External to help differentiate between Internal IP address and external address.

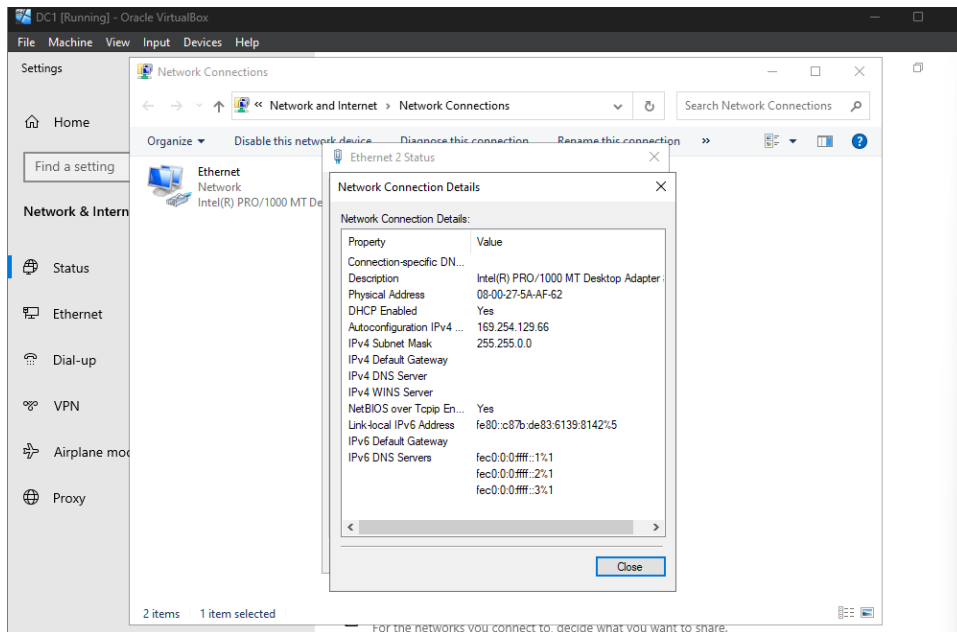


Figure 3. Internal adapter IP address has **Automatic Private IP addressing (APIPA)** (169.254.129.66/ 255.255.0.0) assigned to it, due to failure to obtain a DHCP address.)

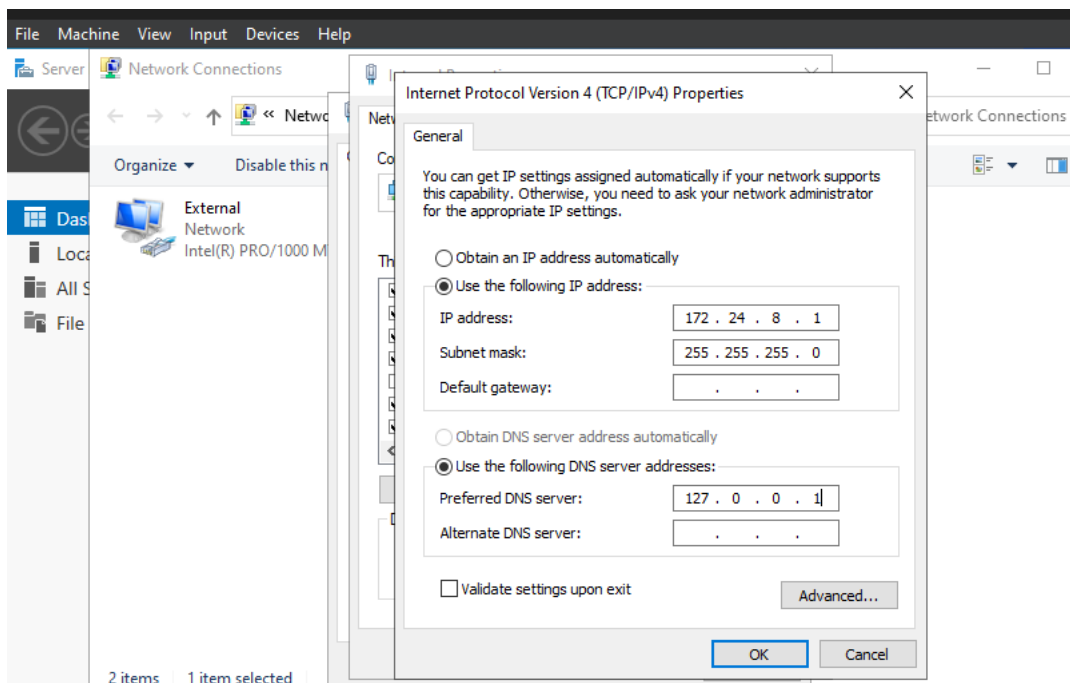


Figure 4. The internal network adapter has been configured with the static IP address 172.24.8.1. This provides a consistent network address that future Windows clients will use to locate the Domain Controller for DNS, DHCP, and Active Directory services.